
A Low-Cost SDR-Based Edge IoT Architecture for Real-Time Decoding of Weather Telemetry in Cyber-Physical Greenhouse Drying Systems

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Abstract: Protected agriculture and solar/greenhouse drying systems require reliable real-time information about external weather conditions to support proactive control decisions. However, professional weather stations often rely on proprietary telemetry interfaces, which may limit their integration with custom cyber-physical systems (CPS). This paper presents a low-cost edge Internet-of-Things (IoT) architecture based on software-defined radio (SDR) for the real-time reception, validation, decoding, and storage of weather telemetry from a Davis Vantage Pro 2 station operating in the 902–928 MHz ISM band. The proposed system was implemented on an ARM/Linux platform using Python and an RTL-SDR receiver. The architecture includes packet validation through CRC-16, bit-level decoding of multiplexed environmental variables, asynchronous data ingestion into a time-series database, local backup logging, and a watchdog-based auto-recovery mechanism. Field tests conducted in Oaxaca, Mexico, over a five-day operating period yielded 7,167 validated database records, an RF packet delivery ratio of 93.23%, a mean time to recovery of 1.03 s after receiver stalls, and a database-writing jitter of 0.3815 s around a 60.18 s average interval. These results indicate that SDR-based edge decoding can provide a reliable data acquisition layer for future closed-loop control of greenhouse fruit-drying systems. The main contribution is not only the decoding of proprietary weather telemetry, but the transformation of such telemetry into a validated, fault-tolerant, and queryable edge data layer for agricultural decision support. The proposed approach contributes an interoperable and reproducible pathway for integrating proprietary meteorological telemetry into agricultural CPS applications.

Keywords: software-defined radio; cyber-physical systems; Internet of Things; precision agriculture; greenhouse drying; edge computing; weather telemetry; reliability engineering

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1 Introduction

The automation of protected agriculture has progressively shifted from isolated sensing systems toward cyber-physical systems (CPS), where embedded computation, communication networks, environmental sensing, and physical actuation are integrated for real-time decision-making [Lee \(2008\)](#); [Gubbi et al. \(2013\)](#); [?](#); [?](#). In greenhouse drying applications, external meteorological variables such as air temperature, relative humidity, precipitation, wind, and solar irradiance are not merely descriptive measurements; they are control inputs that determine whether ventilation, heating, or enclosure actions are beneficial or harmful to the drying process [Mujumdar \(2014\)](#); [Wolfert et al. \(2017\)](#); [?](#); [?](#).

In fruit-drying greenhouses, a purely internal sensing strategy is often insufficient. For example, if the internal relative humidity is high, a reactive controller may open vents or

activate extractors. Nevertheless, if external air is saturated or precipitation is occurring, ventilation could reintroduce moisture and compromise product quality. Therefore, robust access to external meteorological data is essential for proactive control strategies in low-cost drying systems intended for small-scale producers. Greenhouse dryers are particularly attractive for regional agro-industrial applications because they can reduce direct exposure to dust, rainfall, insects, and animals while using solar energy as the main thermal source ???.

Professional weather stations can provide high-quality environmental measurements, but integration with custom controllers may be constrained by proprietary radio protocols or closed commercial interfaces. Software-defined radio (SDR) offers a flexible alternative by replacing part of the radiofrequency signal processing chain with software, enabling low-cost reception and analysis of telemetry signals [Mitola \(1995\)](#); [Collins \(2002\)](#); ?. In this context, the present work investigates the use of SDR as an edge acquisition layer for decoding and storing external weather telemetry in support of a future CPS for greenhouse fruit drying.

The main contribution of this study is the design and field evaluation of a low-cost SDR–edge IoT architecture capable of receiving, validating, decoding, and storing weather telemetry in real time under outdoor operating conditions. Unlike a simple data logger, the proposed architecture includes cyclic redundancy check (CRC) validation, bit-level decoding of multiplexed variables, asynchronous processing, local and time-series database storage, and a watchdog-based auto-recovery mechanism for improving availability on resource-constrained hardware. The architecture therefore contributes to the larger trend of IoT-enabled protected agriculture, where local sensing, edge processing, decision support, and interoperability are increasingly required for practical deployment ???.

The specific objectives were: (i) to implement an SDR-based receiver for weather telemetry in the 902–928 MHz ISM band; (ii) to decode meteorological variables from validated payloads; (iii) to structure the data flow for edge storage and future controller access; and (iv) to quantify reliability using packet delivery ratio, mean time to recovery, writing jitter, and Weibull-based failure behavior.

2 Related Work

2.1 IoT and CPS in protected agriculture

IoT-based monitoring has become a common approach in protected agriculture because it enables continuous acquisition of microclimatic information, remote visualization, and automated decision support [Akyildiz et al. \(2002\)](#); [Gubbi et al. \(2013\)](#); ?; ?; ?. Smart farming studies have highlighted the relevance of sensor networks, data analytics, and distributed decision support for improving agricultural productivity and sustainability [Wolfert et al. \(2017\)](#); [Liakos et al. \(2018\)](#); ?; ?. However, agricultural CPS require more than data visualization: they require reliable data flows, synchronization between physical and digital states, and mechanisms to tolerate faults in field-deployed hardware [Lee \(2008\)](#); [Shi et al. \(2016\)](#); ?; ?. This is particularly important for drying applications, where the process is sensitive to temperature and humidity excursions [Mujumdar \(2014\)](#); ?; ?.

2.2 SDR for environmental telemetry

SDR systems provide a flexible platform for receiving, demodulating, and analyzing radio signals using reconfigurable software workflows [Mitola \(1995\)](#); [Collins \(2002\)](#). Low-cost RTL-SDR devices have been widely adopted in educational, experimental, and monitoring applications because they provide access to radiofrequency signals without requiring dedicated proprietary receivers. For weather telemetry, SDR can enable interoperability when commercial sensors transmit data wirelessly but do not expose open digital interfaces. Community-developed tools such as `rtl-davis` demonstrate that SDR approaches can recover telemetry from Davis weather stations, although additional work is necessary to transform decoded packets into fault-tolerant agricultural CPS data streams [WeeWX community \(2026\)](#); ?.

2.3 Edge computing and time-series data management

Edge computing is well suited to agricultural environments where connectivity may be intermittent and where local decisions must continue even if cloud services are unavailable [Shi et al. \(2016\)](#); ?; ?. Time-series databases are appropriate for environmental monitoring because they organize timestamped observations efficiently and allow downstream applications, dashboards, and controllers to query recent or historical states [InfluxData \(2026\)](#); ?. Therefore, integrating SDR reception with edge validation and time-series storage can support local decision-making in small-scale protected agriculture. This aligns with recent decision-support approaches in agriculture, where sensor data, software platforms, and user-oriented interfaces are combined to guide resource-efficient management decisions ???.

3 Materials and Methods

3.1 Application context

The target application is a mini-greenhouse system for fruit drying in Oaxaca, Mexico. The external weather acquisition layer was designed as the first stage of a CPS in which outside meteorological conditions are used together with internal chamber measurements to support future control of ventilation, heating, and vent-position actuators. Figure ?? summarises the proposed acquisition, validation, storage, and future control workflow.

3.2 Hardware platform

The acquisition system was implemented using a low-cost RTL-SDR receiver connected to an ARM-based single-board computer running Ubuntu Linux on the `aarch64` architecture. The SDR receiver was selected to replace the need for a proprietary console interface and to enable real-time interception of weather telemetry. The receiver used a temperature-compensated crystal oscillator to reduce frequency drift during extended operation.

3.3 Weather telemetry reception

The weather station transmits in the 902–928 MHz ISM band using frequency-hopping spread spectrum (FHSS). The transmitter changes channels approximately every few

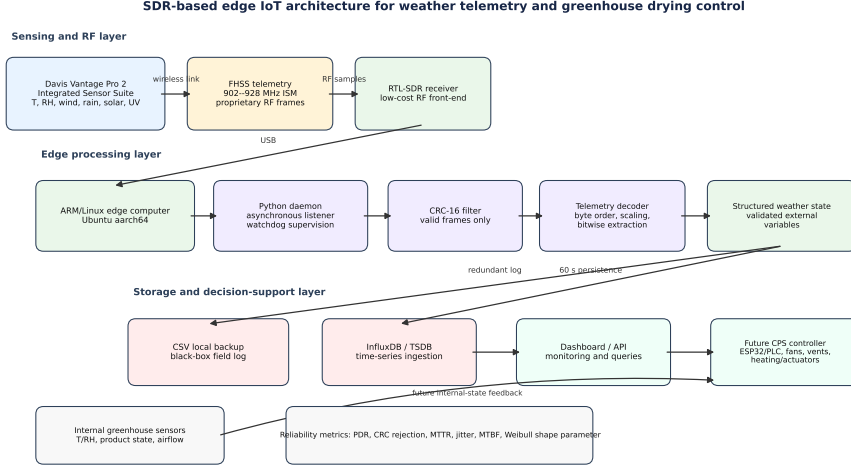


Figure 1 Proposed SDR-based edge IoT architecture for weather-telemetry decoding and future greenhouse drying control. The architecture links the Davis integrated sensor suite and FHSS radio telemetry with an RTL-SDR receiver, an ARM/Linux edge computer, a Python daemon with watchdog supervision, CRC-16 validation, variable decoding, redundant local CSV logging, time-series database ingestion, a dashboard/API layer, and future greenhouse actuators.

seconds. The receiver software was configured to follow the radio activity and extract hexadecimal payloads from valid telemetry bursts.

3.4 Packet structure and CRC validation

Each intercepted packet was treated as an 8-byte frame. The last two bytes were used for CRC-16 validation, whereas the first six bytes contained the payload used for variable decoding. Frames whose computed CRC did not match the received CRC were rejected to avoid propagating corrupted measurements to the database or future control logic.

The packet delivery ratio (PDR) was computed as

$$\text{PDR}(\%) = \frac{N_{\text{valid}}}{N_{\text{valid}} + N_{\text{CRC errors}}} \times 100, \quad (1)$$

where N_{valid} is the number of accepted packets and $N_{\text{CRC errors}}$ is the number of packets rejected by the CRC filter.

3.5 Decoding of meteorological variables

Wind speed and wind direction were present in the fixed portion of the telemetry frame. Wind direction was mapped from an 8-bit integer to degrees using

$$\theta = \frac{x}{255} \times 360, \quad (2)$$

where x is the decoded byte value.

Temperature and relative humidity were decoded from multiplexed payload bytes. Temperature was interpreted as a signed 16-bit value with scaling and unit conversion, whereas relative humidity required fractional byte ordering and 12-bit extraction. Solar irradiance and UV index were decoded using bitwise operations over compact 10-bit fields. The general extraction form used for optical variables was

$$V_{10} = ((b_3 \ll 2) | (b_4 \gg 6)) \& 0x3FF, \quad (3)$$

where b_3 and b_4 are payload bytes, $\ll$ and $\gg$ denote bit shifts, $|$ is bitwise OR, and $\&$ applies a 10-bit mask.

Table ?? summarises the main decoded variables, the telemetry interpretation strategy, and their relevance for greenhouse drying decision support.

Table 1 Decoded weather variables and relevance for greenhouse drying decision support.

Variable	Decoding approach	Relevance for greenhouse drying
Air temperature	Signed 16-bit conversion with scaling and unit transformation	Indicates the external thermal potential available for solar-assisted drying and ventilation decisions.
Relative humidity	Multiplexed 12-bit extraction with fractional byte ordering	Determines whether outside air is likely to remove or introduce moisture into the drying chamber.
Solar irradiance	10-bit bitwise extraction from compact optical payload fields	Indicates the natural solar-energy availability for greenhouse heating and drying.
UV index	10-bit bitwise extraction and scale conversion	Provides an additional optical consistency indicator for daylight and cloud-transient behaviour.
Wind speed and direction	Fixed payload fields decoded from recurrent telemetry frames	Supports risk assessment for ventilation, air exchange, and external exposure conditions.
Rain state/rate	Rain-related payload interpretation and dry-state marker handling	Can inhibit vent opening during rainfall or high-moisture external conditions.

3.6 Software architecture

The software was refactored from a linear prototype into an asynchronous daemon. One thread was dedicated to SDR listening, while the main loop handled screen updates, data formatting, and persistence. Shared data structures were protected using locks to avoid race conditions. Circular buffers were used for short-term trends to prevent unbounded memory growth.

3.7 Data persistence

The data pipeline used two persistence mechanisms. First, a local CSV file was written as a redundant black-box log in case of network or database failure. Second, validated and decoded variables were ingested into a time-series database at approximately 60 s intervals. This structure allows dashboards and future controllers to read the latest external weather state without interrupting SDR reception.

3.8 Console inspection and pending datalogger validation

During field operation, the decoded variables were periodically checked against the values displayed by the original Davis console, and no systematic discrepancies were observed during routine inspection. However, because a Davis datalogger was not available during the reported field test, a fully synchronised quantitative comparison between console logs and SDR-decoded records was not included in this stage. A future datalogger-based validation will align both data streams by timestamp and compute bias, mean absolute error, root mean square error, correlation coefficients, and missing-data percentages for each decoded variable. This future comparison will evaluate the consistency of the decoding workflow with the manufacturer's receiver interpretation; it will not replace metrological calibration against independent reference sensors ?.

3.9 Watchdog and reliability metrics

A watchdog timer was implemented to detect receiver stalls. If no radiofrequency activity was detected during a 30 s interval, the SDR subprocess was terminated and restarted. The mean time to recovery (MTTR) was calculated as

$$\text{MTTR} = \frac{1}{n} \sum_{i=1}^n t_{r,i}, \quad (4)$$

where $t_{r,i}$ is the recovery time of the i -th watchdog intervention.

Database-writing jitter was quantified as the standard deviation of the time intervals between consecutive database writes:

$$J = \sqrt{\frac{\sum_{i=1}^n (\Delta t_i - \bar{\Delta t})^2}{n-1}}. \quad (5)$$

For long-term reliability interpretation, the times between receiver stalls were fitted to a Weibull distribution:

$$f(t) = \frac{\beta}{\eta} \left(\frac{t}{\eta} \right)^{\beta-1} e^{-(t/\eta)^\beta}, \quad (6)$$

where β is the shape parameter and η is the scale parameter [Weibull \(1951\)](#); [O'Connor and Kleyner \(2012\)](#); ?.

4 Results

4.1 Telemetry decoding and data availability

During the field evaluation period from 12 to 16 June 2026, the system stored 7,167 validated database records. The data stream remained available during the test period, and no permanent software failure was observed. This result indicates that the edge architecture was able to maintain continuous acquisition and persistence of decoded weather variables under the evaluated field conditions. Table 1 summarises the main reliability and performance indicators obtained during the test.

Table 2 Summary of reliability and performance metrics obtained during the field evaluation.

Metric	Observed value
Evaluation period	12–16 June 2026
Validated database records	7,167
Estimated intercepted packets	184,508
CRC-rejected packets	12,500
Packet delivery ratio	93.23%
Watchdog interventions	154
Mean time to recovery	1.03 s
Mean database writing interval	60.18 s
Writing jitter	0.3815 s
Empirical MTBF	65.81 min
Weibull shape parameter β	0.3552

4.2 RF link quality

Approximately 184,508 packets were intercepted during the evaluation. Of these, 12,500 frames were rejected as corrupted according to the CRC filter. The resulting PDR was 93.23%, showing that most of the intercepted frames were suitable for decoding despite operation in the unlicensed ISM band.

4.3 Thermo-hygrometric behavior

The decoded temperature and relative humidity series followed the expected inverse daily pattern. As shown in Figure ??, the air-temperature profile increased during daylight hours and decreased during the night and early morning. Figure ?? shows the complementary relative-humidity response, with higher values during cooler periods and a daytime humidity valley that is relevant for identifying favourable drying windows. A 15 min moving average was applied to reduce short-term sensor noise and reveal the macroscopic daily trend.

4.4 Solar irradiance and UV index

The decoded solar irradiance and UV index showed a clear diurnal cycle, with near-zero nighttime values and simultaneous daytime peaks (Figure ??). Short-term drops in both series were observed during daylight hours and are consistent with transient cloud passages. The zero nighttime floor supports the correct masking of the 10-bit optical variables and indicates that neighbouring payload bits did not contaminate the decoded values under dark conditions.

4.5 System recovery and writing stability

The watchdog mechanism was activated 154 times during the evaluation. The mean recovery time was 1.03 s, which is substantially shorter than the 60 s database-writing interval. Therefore, receiver stalls were effectively masked from the persistence layer. The mean writing interval was 60.18 s, with a jitter of 0.3815 s, indicating stable periodic data ingestion and no apparent blocking caused by the multithreaded architecture.

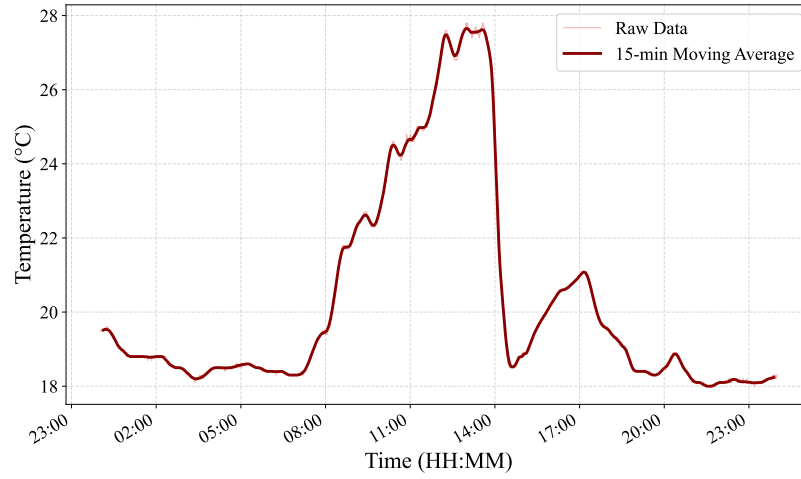


Figure 2 Representative 24 h air-temperature time series obtained from the SDR-based edge telemetry system. The raw decoded measurements are shown together with a 15 min moving average used to highlight the daily thermal trend.

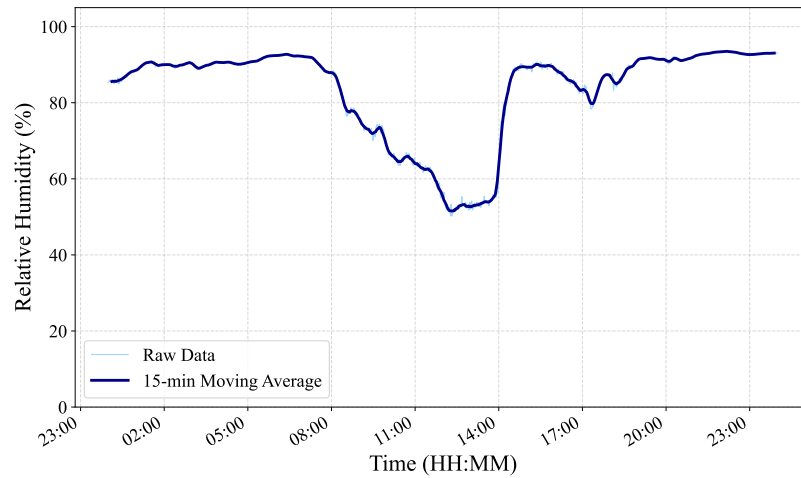


Figure 3 Representative 24 h relative-humidity time series obtained from the SDR-based edge telemetry system. The daytime humidity valley illustrates the type of external condition that can be used to identify favourable ventilation and drying windows.

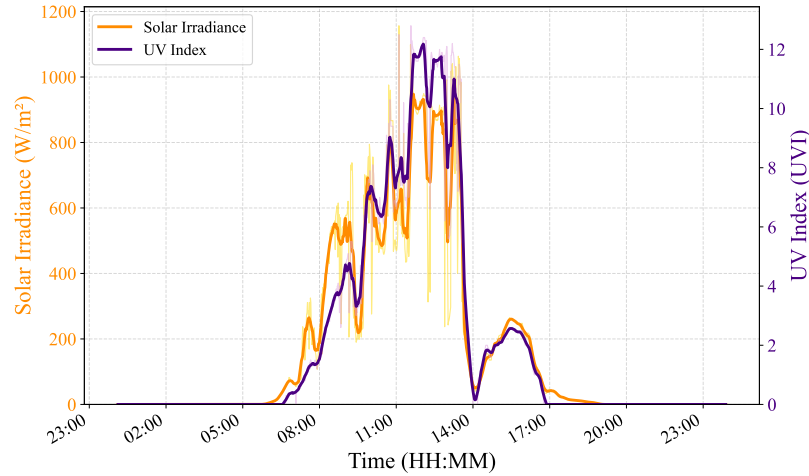


Figure 4 Solar irradiance and UV index profiles obtained from decoded 10-bit payload fields. The similar diurnal behavior and the near-zero nighttime floor support the physical consistency of the optical-variable decoding procedure.

4.6 Weibull reliability interpretation

The empirical MTBF between receiver stalls was 65.81 min. The Weibull shape parameter was $\beta = 0.3552$, which is below unity and suggests a decreasing failure rate pattern commonly associated with early-life or initialization-related failures. In practical terms, the result indicates that the low-cost SDR receiver may experience frequent stalls if used without software recovery. Therefore, the watchdog mechanism is not an optional feature but a key requirement for field deployment.

5 Discussion

The most relevant outcome of this work is not only the reception of weather telemetry through a low-cost SDR device, but the transformation of such telemetry into a validated, persistent, and fault-tolerant edge data layer for agricultural decision support. The proposed architecture can be interpreted as an interoperability layer between a proprietary meteorological telemetry source and a custom agricultural CPS. This is relevant for small-scale protected agriculture because commercial weather stations can provide useful measurements, but their closed communication interfaces may limit integration with local controllers, dashboards, and decision-support routines ???.

From the RF-acquisition perspective, the measured PDR of 93.23% indicates that most intercepted packets were suitable for decoding after CRC filtering. At the same time, the 6.77% CRC-rejection proportion confirms the need for explicit packet validation before any measurement is used by a greenhouse controller. Since drying control typically operates at a slower time scale than the raw RF transmission interval, the loss of individual radio frames does not necessarily compromise the one-minute environmental state required for supervisory decisions. In this context, valid recent data are more important than capturing every transmitted packet.

The watchdog analysis further shows that low-cost SDR hardware should not be treated as an always-stable industrial device under continuous field operation. The empirical MTBF of 65.81 min and the Weibull shape parameter below unity suggest frequent early-type receiver stalls, likely associated with USB-level or thermal behaviour under continuous scanning. However, the MTTR of 1.03 s was much shorter than the 60 s database-writing interval. Therefore, most receiver stalls were absorbed by the software-recovery mechanism before they became persistent gaps in the stored time series. This result supports the inclusion of watchdog supervision as a design requirement for low-cost SDR deployments in agricultural environments, where IoT reliability and resilience are recognised as critical limitations for field applications [O'Connor and Kleyner \(2012\)](#); ?); ?.

From an agricultural management perspective, the decoded variables provide the external environmental context required for proactive greenhouse drying decisions. External relative humidity determines whether ventilation is likely to remove moisture from the drying chamber or introduce humid air. Rain-related information can inhibit vent opening, while solar irradiance can indicate periods of high natural drying potential. Wind speed and wind direction can also support ventilation-risk assessment. Therefore, the proposed system should be interpreted as a decision-support acquisition layer rather than a final closed-loop controller ???.

The visual behaviour of the decoded variables supports the physical consistency of the telemetry-decoding process. The temperature and relative-humidity profiles followed the expected inverse daily relationship, while solar irradiance and UV index showed similar diurnal patterns and a near-zero nighttime floor. These observations are consistent with physically meaningful environmental behaviour and provide qualitative evidence that the byte ordering, scaling, and bitwise extraction routines are operating coherently. Nevertheless, formal quantitative validation against synchronised Davis console logs remains necessary and will be performed when the datalogger is available.

Several limitations should be considered. First, the present evaluation was performed using one weather station, one SDR receiver, and one field location. Second, the reported experiment covered five days, which is sufficient for a proof-of-concept reliability assessment but not for seasonal deployment analysis. Third, the comparison with the original Davis console was performed through periodic visual inspection rather than synchronised datalogger records. Fourth, the study did not include independent calibrated reference sensors, so it validates decoding consistency and system reliability rather than absolute metrological accuracy. Fifth, the greenhouse drying control loop was not closed in this stage; actuator integration and drying-quality metrics remain part of future work.

The architecture is nevertheless reproducible and scalable because it relies on low-cost SDR hardware, an ARM/Linux edge platform, explicit CRC validation, open software components, and standard time-series data storage. Deployment in other locations would require verification of antenna placement, regional radio configuration, environmental interference, and hardware thermal behaviour. In future implementations, the architecture should be combined with internal chamber sensors, actuator feedback, and drying-quality measurements to quantify its effect on drying kinetics, energy use, and product quality ???.

6 Conclusions

This study presented and evaluated a low-cost SDR-based edge IoT architecture for real-time decoding of weather telemetry intended for cyber-physical greenhouse drying systems.

The system achieved reliable acquisition, CRC-based validation, asynchronous processing, redundant storage, and automatic recovery from SDR stalls during multi-day field operation. The measured PDR of 93.23%, MTTR of 1.03 s, and writing jitter of 0.3815 s indicate that the architecture is suitable as an external meteorological data layer for future closed-loop control.

Future work will focus on integrating the decoded external weather data with internal greenhouse sensors and actuator control. Additional validation against calibrated reference sensors, longer endurance tests, and drying experiments with fruit products are required to quantify the impact of the proposed architecture on drying kinetics, product quality, and energy efficiency.

Data and Code Availability

The source code, anonymised logs, processed datasets, and figure-generation scripts will be deposited in a public repository before final submission or made available from the corresponding author upon reasonable request during peer review. The repository will include SDR decoding scripts, CRC validation routines, database schema, sample raw hexadecimal packets, processed CSV files, reliability-analysis scripts, and documentation for reproducing the figures and metrics reported in this paper.

Ethical and Legal Interoperability Statement

The SDR decoding workflow presented in this study was developed for academic research, interoperability, and integration of locally owned environmental sensing equipment into a custom agricultural CPS. The system is intended for passive reception of weather telemetry generated by the authors' own station and is not intended to interfere with, disrupt, or compromise any third-party communication system. Authors should verify local radiofrequency regulations and intellectual-property constraints before field deployment or publication.

Author Contributions

Conceptualization, O.A.H.-F.; methodology, M.U.J.-R., A.R.-S., M.S.C.-M., H.N.-B., D.S.-D. and O.A.H.-F.; software, M.U.J.-R., A.R.-S., H.N.-B. and D.S.-D.; validation, M.U.J.-R., A.R.-S., M.S.C.-M., H.N.-B. and D.S.-D.; formal analysis, O.A.H.-F. and student authors; investigation, all authors; writing—original draft preparation, student authors and O.A.H.-F.; writing—review and editing, O.A.H.-F.; supervision, O.A.H.-F. All authors have read and agreed to the submitted version of the manuscript.

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Conflicts of Interest

The authors declare no conflict of interest.

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